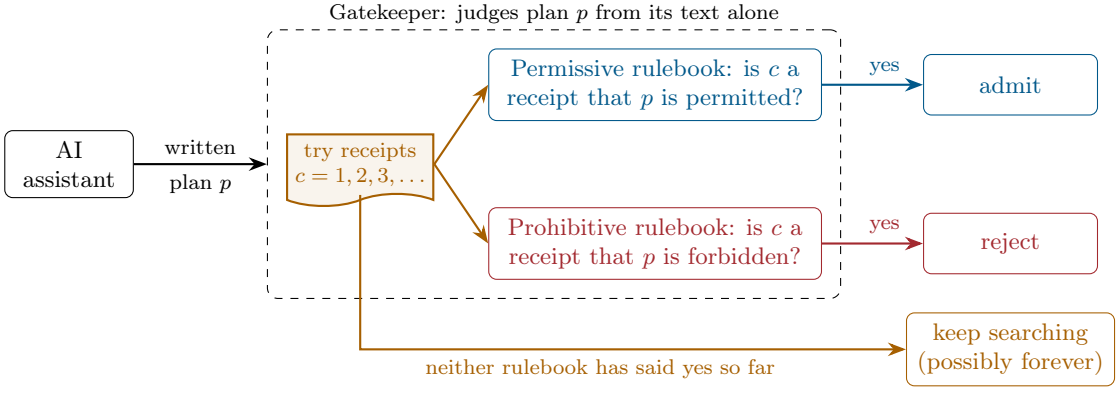
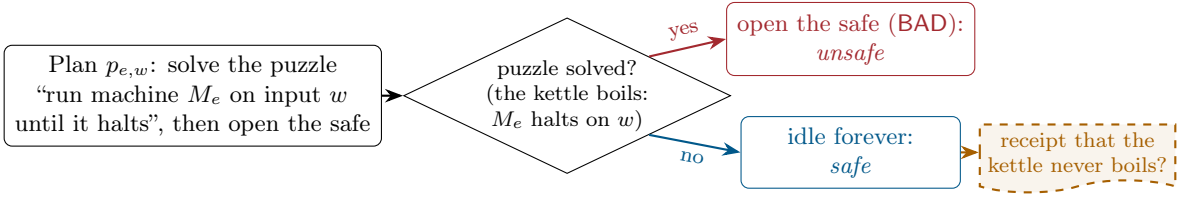
(a) The gatekeeper reads a written plan and looks for receipts in two rulebooks



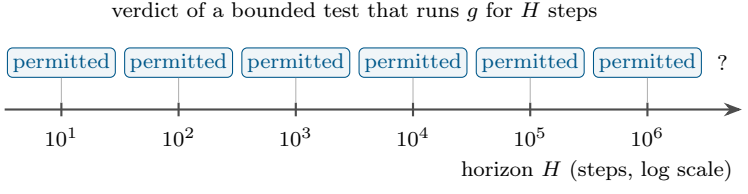
(b) "Solve this puzzle, then open the safe": safe exactly when the puzzle has no solution



A receipt that $p_{e,w}$ is safe would be a receipt that the kettle never boils, and no checkable rulebook has such receipts for every machine M_e and input w. Hence no receipt-checking rulebook admits every safe plan and no unsafe one.

(c) A plan whose safety is an open problem: Goldbach's conjecture

Plan g:
for $n = 4, 6, 8, \ldots$:
  if n is not a sum of two primes,
    open the safe



g is safe exactly when Goldbach's conjecture is true, an open problem. Every bounded test says "permitted"; a receipt that g is safe would settle the conjecture.